Math 220B Midterm

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1 D

Problem 1. Let A be a commutative ring. Prove or disprove that A is a local ring if and only if $A[[x]]$ is a local ring.

Solution. We'll prove that the two statements are equivalent.
Before the actual proof, let's do some observations:

(1) Consider the ideal $(x) \subseteq A[[x]]$ and its effect: Given any power series $f(x) = \sum_{n\geq 0} a_n x^n$, the term $1 - x \cdot f(x) = 1 - \sum_{n\geq 0} a_n x^{n+1}$, which has constant term being 1 (a unit in A), showing that $1 - x \cdot f(x) \in A[[x]]$ is invertible (this is due to the homework problem, where $f(x) = \sum_{n\geq 0} a_n x^n$ is invertible in $A[[x]]$ iff $a_0 \in U(A)$). Hence, by the characterization of Jacobson Radical, $x \in J(A[[x]])$,so the ideal $(x) \subseteq J(A[[x]])$, which (x) is contained in all maximal ideal of $A[[x]]$.

(2) Consider the surjective map $\varphi : A[[x]] \twoheadrightarrow A$ by $\varphi\left(\sum_{n\geq 0} a_n x^n\right) = a_0$. This is in fact a ring homomorphism, because for any power series $f(x) = \sum_{n\geq 0} a_n x^n$ and $g(x) = \sum_{m\geq 0} b_m x^m$ in $A[[x]]$, they satisfy the following three equations:

$$\varphi(f(x) + g(x)) = \varphi\left(\sum_{n\geq 0}(a_n + b_n)x^n\right) = a_0 + b_0 = \varphi(f(x)) + \varphi(g(x)) \tag{1.1}$$

$$\varphi(f(x)g(x)) = \varphi\left(\sum_{k\geq 0}\left(\sum_{n=0}^{k} a_n b_{k-n}\right)x^k\right) = a_0 b_0 = \varphi(f(x))\varphi(g(x)) \tag{1.2}$$

$$\varphi(1) = 1 \in A \tag{1.3}$$

Which, given any power series $f(x) = \sum_{n\geq 0} a_n x^n \in A[[x]]$, one has:

$$\begin{aligned} f(x) \in \ker(\varphi) \quad &\iff \quad \varphi(f(x)) = a_0 = 0 \quad \iff \quad f(x) = \sum_{n\geq 1} a_n x^n \\ &\iff \quad f(x) = x \cdot \sum_{n\geq 1} a_n x^{n-1} \quad \iff \quad f(x) \in (x) \end{aligned} \tag{1.4}$$

This proves that $\ker(\varphi) = (x)$.

Hence, one deduces that $A[[x]]/(x) \cong A$; moreover, the sets $\{\text{ideals } I \subseteq A[[x]] \mid (x) \subseteq I\} \longleftrightarrow \{\text{ideals } J \subseteq A\}$ are in one-to-one correspondance via the associations $I \mapsto \varphi(I)$ and $\varphi^{-1}(J) \mapsfrom J$.

Furthermore, notice that such bijection preserves the inclusion as partial order: Given two ideals $I, I' \subseteq A[[x]]$ containing (x), then $I \subseteq I'$ implies $\varphi(I) \subseteq \varphi(I')$; similarly, given two ideals $J, J' \subseteq A$, if $J \subseteq J'$, one has $\varphi^{-1}(J) \subseteq \varphi^{-1}(J')$.

Now, let's prove the actual statement.

$\Longrightarrow$:

First, suppose A is a local ring, let $\mathfrak{m} \subsetneq A$ denotes its unique maximal ideal, then any proper ideal $J \subsetneq A$ satisfies $J \subseteq \mathfrak{m}$ (since all proper ideal is contained in some maximal ideal, while here there's only one such choice). Hence, for any maximal ideal $M \subsetneq A[[x]]$, since by (1) above we have $(x) \subseteq M$ (which, M is in the set $\{\text{ideal } I \subseteq A[[x]] \mid (x) \subseteq I\}$), then by (2), one has $M = \varphi^{-1}(\varphi(M)) \subseteq \varphi^{-1}(\mathfrak{m})$ (since $\varphi(M)$ is a proper ideal). With $\mathfrak{m} \subsetneq A$ being a proper ideal, $\varphi^{-1}(\mathfrak{m}) \subseteq A[[x]]$ is also proper, so $M \subseteq \varphi^{-1}(\mathfrak{m})$ together with M being a maximal ideal in $A[[x]]$ implies $M = \varphi^{-1}(\mathfrak{m})$.

Therefore, there exists unique maximal ideal in $A[[x]]$, namely $\varphi^{-1}(\mathfrak{m})$, showing $A[[x]]$ is a local ring.

$\Longleftarrow$:

Then, suppose $A[[x]]$ is a local ring, let $M \subsetneq A[[x]]$ denotes its unique maximal ideal. Then, for any maximal ideal $\mathfrak{m} \subsetneq A$, since $\varphi^{-1}(\mathfrak{m})$ is a proper ideal in $A[[x]]$, one has $\varphi^{-1}(\mathfrak{m}) \subseteq M$ (similar argument to previous part). Using the correspondance in (2), one gets $\mathfrak{m} = \varphi(\varphi^{-1}(\mathfrak{m})) \subseteq \varphi(M)$. Yet, with $\varphi(M)$ being a proper ideal (since M is proper), and with $\mathfrak{m} \subsetneq A$ be a maximal ideal, then $\mathfrak{m} = \varphi(M)$.

Thus, there exists unique maximal ideal in A, namely $\varphi(M)$, showing A is a local ring.

The above two proofs conclude that A is a local ring iff $A[[x]]$ is a local ring. $\square$

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Problem 2. Let A be a commutative ring and $a \neq 0$ be a non-unit element of A. Prove or disprove that we can find a multiplicatively closed subset $S \subset A$ such that the image of a under the localization $A \to S^{-1}A$ lies in every maximal ideal of $S^{-1}A$.

Solution. We'll prove that the statement is true.

Since $a \neq 0$ is not a unit, the ideal $(a) \subsetneq A$ is proper, hence there exists a maximal ideal $M \subsetneq A$, where $(a) \subseteq M$.

Now, since M is also a prime ideal (because it's maximal), then take $S = A \setminus M$ a multiplicatively closed set, one has the localization $S^{-1}A = A_M$, while under the map $\varphi : A \to A_M$ by $\varphi(x) = \frac{x}{1}$ has the extension of M, say $M^e = MA_M$ being the unique maximal ideal of A_M (property when doing localization with respect to a prime ideal, it's proven in class).

Then, since $a \in M$, one has $\varphi(a) = \frac{a}{1} \in MA_M$ (the unique maximal ideal of $S^{-1}A = A_M$). Hence, the image of a under this localization lies in every maximal ideal of $S^{-1}A$ (since there's only one maximal ideal). This is a desired result for the statement.

$\square$

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Problem 3. Let A be a finite integral domain. Prove or disprove that A is a quotient of $\mathbb{Z}$.

Solution. We'll prove that A is not necessarily a quotient of $\mathbb{Z}$, by constructing a counterexample.

Consider the field $k = \mathbb{Z}/2\mathbb{Z}$, the polynomial ring $k[x]$ (an integral domain), and the polynomial $x^2 + x + 1 \in k[x]$, we'll use these to construct a counterexample.

Given that $0, 1$ are the only elements in $k = \mathbb{Z}/2\mathbb{Z}$, if plugin $x^2 + x + 1$, one gets:

$$0^2 + 0 + 1 = 1 \neq 0, \quad 1^2 + 1 + 1 = 1 + 1 + 1 = 1 \neq 0 \tag{3.1}$$

This shows that $x^2 + x + 1$ has no solution in k; also, because it is of degree 2, it can't be factored into any smaller degree nonconstant polynomial (since smaller degree nonconstant polynomials are degree 1, which always has a solution in a field), so $x^2 + x + 1 \in k[x]$ is an irreducible element. Because $k[x]$ is a PID (since k is a field), then $(x^2 + x + 1) \subset k[x]$ is a maximal ideal, the ring $A = k[x]/(x^2 + x + 1)$ is a field (in particular an integral domain).

Now, we claim that A is a finite integral domain, and is not a quotient of $\mathbb{Z}$:

(1) We first claim that $A = \left\{\overline{a} + \overline{bx} \mid a, b \in k\right\}$ (which since $k = \mathbb{Z}/2\mathbb{Z}$ is finite, this implies A is finite).

Given x^2, since one has $\overline{x^2 + x + 1} = 0$ in A, then $\overline{x^2} = \overline{-1 - x} = \overline{1} + \overline{x}$ in A. Now, if consider any degree $n \geq 2$ monomial x^n, since $x^n = x^2 \cdot x^{n-2}$, then one has the following:

$$\overline{x^n} = \overline{x^2} \cdot \overline{x^{n-2}} = \left(\overline{1 + x}\right) \cdot \overline{x^{n-2}} = \overline{x^{n-2}} + \overline{x^{n-1}} \tag{3.2}$$

Which, using induction, since $\overline{x^n}$ can be written as linear combinations of smaller degree monomials, it can be written as linear combinations of $\overline{1}$ and $\overline{x}$.

Hence, since all monimials' quotient can be written in the form $\overline{a} + \overline{bx}$ in A (and all polynomials are finite linear combinations of monomials), all polynomials' quotient can be written as $\overline{a} + \overline{bx}$ for some $a, b \in k$, and one concludes that $A = \left\{\overline{a} + \overline{bx} \mid a, b \in k\right\}$.

Moreover, such representative is unique (which suffices to show the case for $\overline{0}$): Suppose $\overline{a} + \overline{bx} = \overline{0}$ in A, then $a + bx \in (x^2 + x + 1)$ in $k[x]$. But, this enforces $a + bx = 0$ (or else if it's nonzero, then $a + bx = g \cdot (x^2 + x + 1)$ for some $g \in k[x]$; since $\deg(a + bx) = 0$ or 1, while $\deg(g \cdot (x^2 + x + 1)) \geq \deg(x^2 + x + 1) = 2$, this forms a contradiction). Hence, $\overline{a} + \overline{bx} = 0$ enforces $a + bx = 0$, which further enforces $a, b = 0$. So, such representative is unique for each element.

(2) Based on (1), we know $A = \{\overline{0},\ \overline{1},\ \overline{x},\ \overline{1} + \overline{x}\}$ as sets (so A has cardinality 4). In particular, $\overline{1} \neq 0$, and $\overline{1} + \overline{1} = \overline{0}$ (since under $k = \mathbb{Z}/2\mathbb{Z}$, $1 + 1 = 0$), so $\operatorname{char}(A) = 2$.

However, among all the quotient of $\mathbb{Z}$, since the only ideals of $\mathbb{Z}$ are of the form $I = n\mathbb{Z}$ (where $n \in \mathbb{Z}$; WLOG say $n \geq 0$, since if $n < 0$, take $-n > 0$, one has $n\mathbb{Z} = -n\mathbb{Z}$). Which, $\operatorname{char}(\mathbb{Z}/n\mathbb{Z}) = n$, since $\sum_{i=1}^{n} \overline{1} = \overline{n} = \overline{0}$ in $\mathbb{Z}/n\mathbb{Z}$, and for any $0 < k < n$, by definition $\sum_{i=1}^{k} \overline{1} = \overline{k} \neq 0$ (since $k \notin n\mathbb{Z}$).

Now, suppose the contrary that A is a quotient of $\mathbb{Z}$, say $I = n\mathbb{Z}$ is an ideal of $\mathbb{Z}$ satisfying $\mathbb{Z}/n\mathbb{Z} = A$, then $2 = \operatorname{char}(A) = \operatorname{char}(\mathbb{Z}/n\mathbb{Z}) = n$, showing $I = 2\mathbb{Z}$. Yet, this implies $A = \mathbb{Z}/2\mathbb{Z} = \{0,\ 1\}$, or A has cardinality 2 instead of 4, which is a contradiction. Therefore, A cannot be a quotient of $\mathbb{Z}$.

Here, the provided example $A = k[x]/(x^2 + x + 1)$ (for $k = \mathbb{Z}/2\mathbb{Z}$) is an example of a finite integral domain, which is not a quotient of $\mathbb{Z}$, hence a counterexample of the statement.

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Problem 4. Let A be a commutative ring. Prove or disprove that the canonical map $\frac{A[X,Y]}{(XY)} \to A[X] \times A[Y]$, given by $f(X,Y) \mapsto (f(X,0), f(0,Y))$, is an isomorphism of rings.

Solution. We'll prove that it's not an isomorphism of rings, specifically by proving it's not surjective.

First, given the map $\overline{\varphi} : \frac{A[X,Y]}{(XY)} \to A[X] \times A[Y]$ by $\overline{\varphi}(f(X,Y)) = (f(X,0), f(0,Y))$, with the projection $\pi : A[X,Y] \twoheadrightarrow \frac{A[X,Y]}{(XY)}$, one produces a map $\varphi := \overline{\varphi} \circ \pi : A[X,Y] \to A[X] \times A[Y]$ (given by $\varphi(f(X,Y)) = (f(X,0), f(0,Y))$ also, notation wise). Which, because π is surjective, one has $\overline{\varphi}$ being surjective $\iff \varphi$ is surjective. So, the goal is proving φ is not surjective.

Given any $f(X,Y) \in A[X,Y]$, if view this ring as the form $(A[X])[Y]$, one can rewrite $f(X,Y)$ as the following:

$$f(X,Y) = f_n(X)Y^n + ... + f_1(X)Y + f_0(X), \qquad \forall i \in \{0,1,...,n\}, \quad f_i(X) \in A[X] \tag{4.1}$$

Which, plugin to φ, one yields the following:

$$\varphi(f(X,Y)) = (f_0(X), \quad f_n(0)Y^n + ... + f_1(0)Y + f_0(0)) \tag{4.2}$$

Now, we claim that the element $(0,1) \in A[X] \times A[Y]$ is not in the image of φ: Suppose the contrary that $(0,1) \in \mathrm{im}(\varphi)$, there exists $f(X,Y) \in A[X,Y]$ such that $\varphi(f(X,Y)) = (0,1)$. Using the above form, one yields the following:

$$(0,1) = \varphi(f(X,Y)) = (f_0(X), \quad f_n(0)Y^n + ... + f_1(0)Y + f_0(0)) \tag{4.3}$$

In particular, $f_0(X) = 0$ in $A[X]$, while $f_n(0)Y^n + ... + f_1(0)Y + f_0(0) = 1$ in $A[Y]$. Take the second equality, the constant term of $f_n(0)Y^n + ... + f_1(0)Y + f_0(0)$ is $f_0(0)$, showing $f_0(0) = 1$; yet, this contradicts the other equality $f_0(X) = 0$ (which states $f_0(0) = 0$).

So, the element $(0,1) \notin \mathrm{im}(\varphi)$, showing φ is not surjective. This proves that $\frac{A[X,Y]}{(XY)} \to A[X] \times A[Y]$ by $f(X,Y) \mapsto (f(X,0), f(0,Y))$ cannot be an isomorphism.

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Problem 5. Let A be a commutative ring and $B = A[X_1, ..., X_n]/I$ for some maximal ideal $I \subset A[X_1, ..., X_n]$. Prove or disprove that under the canonical composite ring homomorphism $A \to A[X_1, ..., X_n] \twoheadrightarrow B$, B is a finitely generated A-module (i.e., generated by finitely many elements as an A-module).

Solution. We'll prove that B is not necessarily a finitely generated A-module, by constructing a counterexample.

First, let's pick a prime number $p \in \mathbb{Z}$, consider $A = \left\{\frac{n}{m} \in \mathbb{Q} \mid m \notin p\mathbb{Z}\right\}$ (where $\frac{n}{m}$ is assumed to be a reduced fraction). Let's verify A is a ring:

- Since $0 = \frac{0}{1}$ has $1 \notin p\mathbb{Z}$, $0 \in A$ (A has zero), and since $1 = \frac{1}{1}$ has $1 \notin p\mathbb{Z}$, $1 \in A$ (A has multiplicative identity of $\mathbb{Q}$).
- Given $\frac{n_1}{m_1}, \frac{n_2}{m_2} \in A$, since $m_1, m_2 \notin p\mathbb{Z}$, so $m_1 m_2 \notin p\mathbb{Z}$, hence one has $\frac{n_1}{m_1} + \frac{n_2}{m_2} = \frac{n_1 m_2 + n_2 m_1}{m_1 m_2} \in A$ (A is closed under addition of $\mathbb{Q}$).
- Similarly, the above statement ($m_1 m_2 \notin p\mathbb{Z}$) also implies $\frac{n_1}{m_1} \cdot \frac{n_2}{m_2} = \frac{n_1 n_2}{m_1 m_2} \in A$ (A is closed under multiplication of $\mathbb{Q}$).

So, since A contains both $0, 1$, and is closed under $\mathbb{Q}$'s addition and multiplication, it's a subring of $\mathbb{Q}$.

Now, consider the polynomial ring $A[x]$, and an ideal $(px - 1) \subset A[x]$. We aim to use these to construct the desired counterexample.

Since there's a canonical inclusion $A \hookrightarrow \mathbb{Q}$, take the element $\frac{1}{p} \in \mathbb{Q}$, there is a canonical evaluation map $\varphi : A[x] \to \mathbb{Q}$ by $\varphi(x) = \frac{1}{p}$, and $\varphi(r) = r$ for all $r \in A$. Let's deduce some property of φ:

- First, notice that φ is surjective, since given any $\frac{n}{m} \in \mathbb{Q}$ (for $n, m \in \mathbb{Z}$, $m \neq 0$). Which, using the unique factorization property of $\mathbb{Z}$, one can express $m = m_1 \cdot p^k$ for some integer $k \geq 0$, and $m_1 \notin p\mathbb{Z}$. Hence, one has $\frac{n}{m_1} \in A$, and it yields the following:

$$\varphi\left(\frac{n}{m_1} x^k\right) = \frac{n}{m_1} \cdot \frac{1}{p^k} = \frac{n}{m_1 p^k} = \frac{n}{m} \tag{5.1}$$

  This shows the surjectivity of φ.

- Now, if consider the expression $px - 1 \in A[x]$, notice that $\varphi(px - 1) = p \cdot \frac{1}{p} - 1 = 1 - 1 = 0$, so $px - 1 \in \ker(\varphi)$, or $(px - 1) \subseteq \ker(\varphi)$.

  Now, we aim to show that $(px - 1) = \ker(\varphi)$: Supppose $f(x) \in \ker(\varphi)$, then one has $\varphi(f) = f\left(\frac{1}{p}\right) = 0$, showing $\frac{1}{p}$ is a root of $f(x) \in A[x] \hookrightarrow \mathbb{Q}[x]$ (as a polynomial of rational coefficients). Hence, within $\mathbb{Q}[x]$, one has the following factorization:

$$\begin{aligned} &f(x) = \left(x - \frac{1}{p}\right)\left(\frac{m_k}{n_k} x^k + ... + \frac{m_0}{n_0}\right) \\ &\forall 0 \leq i \leq k, \quad \frac{m_i}{n_i} \in \mathbb{Q}, \quad m_i = 0 \text{ or } \gcd(m_i, n_i) = 1 \end{aligned} \tag{5.2}$$

  If expand $f(x)$, one yields back the following:

$$\begin{aligned} f(x) &= \left(\frac{m_k}{n_k} x^{k+1} + ... + \frac{m_0}{n_0} x\right) - \left(\frac{m_k}{p n_k} x^k + ... + \frac{m_0}{p n_0}\right) \\ &= \frac{m_k}{n_k} x^{k+1} + \sum_{i=1}^{k} \left(\frac{m_{i-1}}{n_{i-1}} - \frac{m_i}{p n_i}\right) x^i + \frac{m_0}{p n_0} \in A[x] \quad \text{(After reducing each fraction)} \end{aligned} \tag{5.3}$$

We'll use induction to show that all $m_i \in p\mathbb{Z}$, and $n_i \notin p\mathbb{Z}$:

First, for constant coefficient, one has $\frac{m_0}{pn_0} \in A$. If $m_0 = 0 \in p\mathbb{Z}$ then we're done (since $n_0 = 1 \notin p\mathbb{Z}$ can be chosen). Suppose $m_0 \neq 0$, write $m_0 = m_0' \cdot p^{r_0}$ and $n_0 = n_0' \cdot p^{s_0}$ for integers $m_0', n_0' \notin p\mathbb{Z}$, and $r_0, s_0 \in \mathbb{N}$. With the assumption that $\gcd(m_0, n_0) = 1$, one can't have $r_0, s_0 > 0$ simultaneously (or else p divides both m_0, n_0 is a contradiction); similarly, one can't have $r_0 = 0$ either (or else $\frac{m_0}{pn_0} = \frac{m_0'}{n_0' \cdot p^{s_0+1}}$ has p-factor in the denominator that can't be cancelled, violating the condition for R). Hence, one must have $s_0 = 0$, causing the following:

$$\frac{m_0}{pn_0} = \frac{m_0' \cdot p^{r_0}}{n_0' \cdot p} = \frac{m_0'}{n_0'} \cdot p^{r_0 - 1} \in A \tag{5.4}$$

Because $m_0', n_0' \notin p\mathbb{Z}$, one has $\frac{n_0'}{m_0'} \in A$ (since $m_0' \neq 0$), showing $p^{r_0-1} \in A$. As a result, one must have $r_0 - 1 \geq 0$, or $r_0 \geq 1$, showing that $m_0 \in p\mathbb{Z}$ (and $n_0 \notin p\mathbb{Z}$, because $\gcd(m_0, n_0) = 1$, so $p \mid m_0$ implies $p \nmid n_0$).

Now, inductively suppose for given index $1 \leq i \leq k$, one has $m_{i-1} \in p\mathbb{Z}$ and $n_{i-1} \notin p\mathbb{Z}$. Then, using one of the coefficients of $f(x)$, one gets the following:

$$\frac{m_{i-1}}{n_{i-1}} - \frac{m_i}{pn_i} = \frac{pn_i m_{i-1} - n_{i-1} m_i}{pn_{i-1} n_i} \in A \tag{5.5}$$

Here, if $m_i = 0 \in p\mathbb{Z}$ then we're done (since again $n_i = 1$ can be chosen). So, suppose $m_i \neq 0$, write $m_i = m_i' \cdot p^{r_i}$ and $n_i = n_i' \cdot p^{s_i}$ for integers $m_i', n_i' \notin p\mathbb{Z}$ and $r_i, s_i \in \mathbb{N}$. Again, since $\gcd(m_i, n_i) = 1$, one can't have $r_i, s_i > 0$ simultaneously (or else p divides both m_i, n_i is a contradiction). Similarly, one can't have $r_i = 0$, since then $m_i \notin p\mathbb{Z}$, together with $n_{i-1} \notin p\mathbb{Z}$ (by induction hypothesis), one has $n_{i-1} m_i \notin p\mathbb{Z}$, hence $pn_i m_{i-1} - n_{i-1} m_i \notin p\mathbb{Z}$. Yet, this is a contradiction, since the above term (5.5) has p-factor in the denominator that can't be cancelled out in this case. So, one has $s_i = 0$ again.

Finally, if expand the above terms, we get:

$$\frac{pn_i m_{i-1} - n_{i-1} m_i}{pn_{i-1} n_i} = \frac{n_i' m_{i-1} - n_{i-1} m_i' p^{r_i - 1}}{n_{i-1} n_i'} \in A \tag{5.6}$$

With $n_{i-1}, n_i' \notin p\mathbb{Z}$, the term $\frac{n_i' m_{i-1}}{n_{i-1} n_i'} \in A$; then, one requires $\frac{n_{i-1} m_i' \cdot p^{r_i - 1}}{n_{i-1} n_i'} \in A$, or $p^{r_i - 1} \in A$ (since $n_{i-1}, n_i', n_{i-1}, m_i' \notin p\mathbb{Z}$ are all invertible in A). Then, one must have $r_i - 1 \geq 0$, or $r_i \geq 1$, showing $m_i \in p\mathbb{Z}$. This completes the induction.

As a result, since all $m_i \in p\mathbb{Z}$ and $n_i \notin p\mathbb{Z}$ (or, $m_i = p \cdot m_i'$ for some $m_i' \in \mathbb{Z}$), $f(x)$ can be rewritten as the following:

$$\begin{aligned} f(x) &= \left(x - \frac{1}{p}\right)\left(\frac{m_k}{n_k} x^k + \dots + \frac{m_0}{n_0}\right) \\ &= \left(x - \frac{1}{p}\right) \cdot p \left(\frac{m_k'}{n_k} x^k + \dots + \frac{m_0'}{n_0}\right) \\ &= (px - 1)\left(\frac{m_k'}{n_k} x^k + \dots + \frac{m_0'}{n_0}\right) \end{aligned} \tag{5.7}$$

With each $\frac{m_i'}{n_i} \in R$ (since $n_i \notin p\mathbb{Z}$), one has $f(x) \in (px - 1)$. This shows that $\ker(\varphi) \subseteq (px - 1)$, finishing the proof that $\ker(\varphi) = (px - 1)$.

As a result, the evaluation $\varphi : A[x] \twoheadrightarrow \mathbb{Q}$ by $\varphi(x) = \frac{1}{p}$ is surjective, and has $\ker(\varphi) = (px - 1)$, showing $A[x]/(px - 1) \cong \mathbb{Q}$ (or $(px - 1) \subset A[x]$ is maximal).

Finally, take the composition map $A \hookrightarrow A[x] \twoheadrightarrow A[x]/(px-1) \cong \mathbb{Q}$, it is equivalent to the composition $\varphi \circ \iota : A \to \mathbb{Q}$, which all $r \in A$ has $\varphi \circ \iota(r) = \varphi(r) = r$, being a canonical inclusion map (since $r \in (px-1)$ implies $r = g \cdot (px-1)$ for some $g \in A[x]$, which must have $r = 0$ or $\deg(r) \geq 1$; yet, r being a constant can't have $\deg(r) \geq 1$, so it must be 0).

Here, we claim that $\mathbb{Q}$ is not a finitely generated A-module:

Suppose the contrary that $\mathbb{Q}$ is a finitely-generated A-module under the inclusion $\iota : A \hookrightarrow \mathbb{Q}$, then there exists $\frac{m_1}{n_1}, ..., \frac{m_k}{n_k} \in \mathbb{Q}$, such that $\mathbb{Q} = A\langle \frac{m_1}{n_1}, ..., \frac{m_k}{n_k} \rangle$. Which, for each n_i, one can write $n_i = n_i' \cdot p^{s_i}$ for some $n_i' \notin p\mathbb{Z}$ and $s_i \in \mathbb{N}$. Notice then $\mathbb{Q} = A\langle \frac{m_1}{p^{s_1}}, ..., \frac{m_k}{p^{s_k}} \rangle$, because every $\ell \in \mathbb{Q}$ has the following for some $a_1, ..., a_n \in A$:

$$\ell = \sum_{i=1}^{k} a_i \cdot \frac{m_i}{n_i} = \sum_{i=1} a_i \cdot \frac{m_i}{n_i' \cdot p^{s_i}} = \sum_{i=1}^{k} \frac{a_i}{n_i'} \cdot \frac{m_i}{p^{s_i}}, \qquad \forall i \in \{1, ..., k\}, \quad \frac{a_i}{n_i'} \in A \tag{5.8}$$

This shows that $\left\{\frac{m_1}{n_1}, ..., \frac{m_k}{n_k}\right\}$ and $\left\{\frac{m_1}{p^{s_1}}, ..., \frac{m_k}{p^{s_k}}\right\}$ both generate $\mathbb{Q}$ as A-module.

Yet, notice that for $s = \max\{s_1, .., s_k\}$, the element $\frac{1}{p^{s+1}}$ can't be generated by $\left\{\frac{m_1}{p^{s_1}}, ..., \frac{m_k}{p^{s_k}}\right\}$: Suppose $\frac{1}{p^{s+1}}$ can be generated, there exists some $\frac{a_i}{b_i} \in A$ satisfy the following (with each $b_i \notin p\mathbb{Z}$):

$$\begin{aligned} \frac{1}{p^{s+1}} &= \sum_{i=1}^{k} \frac{a_i}{b_i} \cdot \frac{m_i}{p^{s_i}} = \sum_{i=1}^{k} \frac{a_i m_i \cdot (b_1 ... b_{i-1} b_{i+1} ... b_k) \cdot p^{s-s_i+1}}{b_1 ... b_k \cdot p^{s+1}} \\ \implies b_1 ... b_k &= \sum_{i=1}^{k} a_i m_i \cdot (b_1 ... b_{i-1} b_{i+1} ... b_k) \cdot p^{s-s_i+1} \end{aligned} \tag{5.9}$$

Notice that the right hand side has each term in the sum being divisible by p (since $s - s_i + 1 > 0$ for all index i). Yet, the left hand side can't be divisible by p (since we assume each $b_i \notin p\mathbb{Z}$, so $b_1 ... b_k \notin p\mathbb{Z}$ by the prime property). Hence, we reach a contradiction. This implies our assumption is false, showing $\mathbb{Q} \cong A[x]/(px-1)$ can't be a finitely generated A-module.

As a result, the above A and $B = A[x]/(px-1)$ is an example where $A \hookrightarrow R[x] \twoheadrightarrow B$ has B not being a finitely-generated A-module, which is a desired counterexample for the statement.